

Simple Linear Regression Models

STAT 24510-30040

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Linear regression models is the most commonly used statistical model.

We start with simple linear models with response variable Y and one explanatory variable X .

Data Measurements $(x_i, y_i), i = 1, 2, \dots, n$, are n -pairs of independent observations.

Goal Model (or formulate) the relationship of x_i 's and y_i 's, possibly predict future observations of y given x .

Bivariate Model vs. Conditional Model

(M1) Model the pairs as joint normal (Review)

Under this model, the observed data $(x_i, y_i), i = 1, 2, \dots, n$ are treated as n independent observations of bivariate normal random variable pairs $(X_i, Y_i), i = 1, 2, \dots, n$, which are independent random pairs (vectors) from a common bivariate normal distribution.

$$(X_i, Y_i) \text{ i.i.d. } \sim N_2 \left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{bmatrix} \right), \quad i = 1, \dots, n.$$

The set of five parameters $\{\mu_X, \sigma_X^2, \mu_Y, \sigma_Y^2, \rho\}$ can be estimated by the maximum likelihood method.

The resulting ML estimators are familiar ones for the mean and variance parameters,

$$\hat{\mu}_X = \bar{X}, \quad \hat{\mu}_Y = \bar{Y}, \quad \hat{\sigma}_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \hat{\sigma}_Y^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

and the correlation parameter

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Often we are interested in estimating Y at a given $X = x$ (or estimating X at a given $Y = y$). The joint normal property implies that, conditioned on $X = x$, $Y|X = x$ is also of normal distribution, with conditional mean and conditional variance as

$$\mathbb{E}(Y|X = x) = \mu_Y + \frac{\sigma_Y}{\sigma_X} \rho (x - \mu_X), \quad \text{Var}(Y|X = x) = \sigma_Y^2 (1 - \rho^2)$$

(M2) Model the data by conditioning on $X = x$

If all we are interested is predict Y for given $X = x$ based on observed data $(x_i, y_i), i = 1, 2, \dots, n$, we may model Y_i as independent normal random variable conditioned on observed value $X = x_i$.

$$Y_1, \dots, Y_n \text{ are independent, } \quad Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2) \text{ for given } X = x_i.$$

Equivalently, the above can be written as, for given x_i 's,

$$Y_i = \beta_0 + \beta_1 x_i + \sigma Z_i, \quad Z_i \text{ i.i.d. } \sim N(0, 1), \quad i = 1, \dots, n.$$

This is a linear regression model, commonly written as, for given x_i 's,

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n,$$

which is shorthand for a set of n model equations:

$$\begin{cases} Y_1 &= \beta_0 + \beta_1 x_1 + \varepsilon_1 \\ Y_2 &= \beta_0 + \beta_1 x_2 + \varepsilon_2 \\ &\vdots \\ Y_n &= \beta_0 + \beta_1 x_n + \varepsilon_n \end{cases}$$

By definition of conditional expectation, , for given $X = x$,

$$\mathbb{E}(Y|X = x) = \beta_0 + \beta_1 x, \quad \text{Var}(Y|X = x) = \sigma^2$$

Remarks on the bivariate model (M1) and the conditional model (M2)

- The conditional model (M2) is also called the simple linear regression model.
- In the conditional model (M2), $\mathbb{E}(Y|X = x) = \beta_0 + \beta_1 x$, variable x is treated as deterministic.
- Commonly the conditional part in $\mathbb{E}(Y|X = x)$ and $\text{Var}(Y|X = x)$ is left out, the simplified form becomes

$$\mathbb{E}(Y) = \beta_0 + \beta_1 x, \quad \text{Var}(Y) = \sigma^2$$

However we should remember that linear regression model is based on the conditional expectation assumption.

- If (X, Y) are indeed bivariate normal, from the conditional mean formula $\mathbb{E}(Y|X = x) = \mu_Y + \frac{\sigma_Y}{\sigma_X} \rho (x - \mu_X)$ and the conditional variance formula $\text{Var}(Y|X = x) = \sigma_Y^2 (1 - \rho^2)$, we can derive the relations between the parameters in the two models as

$$\begin{cases} \beta_1 &= \rho \frac{\sigma_Y}{\sigma_X}, \\ \beta_0 &= \mu_Y - \beta_1 \mu_X = \mu_Y - \rho \frac{\sigma_Y}{\sigma_X} \mu_X, \\ \sigma^2 &= \sigma_Y^2 (1 - \rho^2). \end{cases}$$

- Bivariate model in (M1) has stronger assumption than the conditioning linear model in (M2). Model in (M2) does not need to assume the marginal distribution of X .
- If the main goal is to estimate Y for given $X = x$ (as linear regression model does), then we focus on modeling Y conditioned on X , the linear regression way, as in model (M2).

In the following, we focus on the linear model conditioned on fixed, given values $x_1, \dots, x_n$.

Least Squares Estimation for coefficients of the linear model

We start by considering the linear model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim (0, \sigma^2), \quad i = 1, \dots, n,$$

The notation means $\mathbb{E}(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, without imposing the normality assumption.

LS estimators for β_0, β_1

The Least Squares (LS) estimators $\hat{\beta}_0, \hat{\beta}_1$ for β_0, β_1 respectively are chosen to minimize the sum of squared errors, a.k.a. the residual sums of squares (RSS).

$$\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2$$

The optimal solution can be expressed as

$$\{\hat{\beta}_0, \hat{\beta}_1\} = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2$$

where the minimum is among all $(\beta_0, \beta_1) \in \mathbb{R}^2$.

To find the minimizing parameter values, we may take partial derivatives of the sum of squared errors with respect to each of β_1, β_0 , and set the derivatives to zero.

$$\begin{cases} \frac{\partial}{\partial \beta_0} \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2 = -2 \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i)) & = 0 \\ \frac{\partial}{\partial \beta_1} \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2 = -2 \sum_{i=1}^n x_i (Y_i - (\beta_0 + \beta_1 x_i)) & = 0 \end{cases} \quad (1)$$

This gives us two equations with two unknowns β_0, β_1 .

We may write y_i for observed data of Y_i , and use y_i in the place of Y_i for convenience in notation during derivation.

Denote

$$\overline{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i, \quad \overline{x^2} = \frac{1}{n} \sum_{i=1}^n x_i^2$$

the two equations with two unknowns β_0, β_1 in (1) can be re-expressed and then solved,

$$\begin{cases} \beta_0 + \beta_1 \bar{x} - \bar{y} & = 0 \\ \bar{x} \beta_0 + \beta_1 \overline{x^2} - \overline{xy} & = 0 \end{cases}$$

The solutions can be expressed as

$$\begin{cases} \hat{\beta}_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$

The unique pair of critical point is indeed the global minimum, which can be verified, by the differentiability of the RSS with respect to the parameters β_0 and β_1 , the uniqueness of the critical point, and by the boundary properties of the parameter space.

Expressions and interpretations for $\hat{\beta}_0, \hat{\beta}_1$

The solutions in the above form have intuitive interpretations.

- The estimated slope $\hat{\beta}_1$ is in the form of $\frac{\text{sample covariance between } x \text{ and } y}{\text{sample variance of } x}$.
- Since $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, the center of the data $(\bar{x}, \bar{y})$ is on the model predicted line $y = \hat{\beta}_0 + \hat{\beta}_1 x$.

There are several useful expressions for the numerator and denominator in $\hat{\beta}_1$. For the denominator,

$$n(\overline{x^2} - \bar{x}^2) = \sum_{i=1}^n x_i^2 - n\bar{x}^2 = \sum_{i=1}^n (x_i - \bar{x})x_i = \sum_{i=1}^n (x_i - \bar{x})^2$$

And for the numerator,

$$n(\overline{xy} - \bar{x}\bar{y}) = \sum_{i=1}^n x_i y_i - \bar{x}(\sum_{i=1}^n y_i) = \sum_{i=1}^n (x_i - \bar{x})y_i = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

In the above derivation, we apply the fact that $\sum_{i=1}^n (x_i - \bar{x})\bar{x} = 0$, $\sum_{i=1}^n (x_i - \bar{x})\bar{y} = 0$, since $\sum_{i=1}^n (x_i - \bar{x}) = 0$.

The resulting unique LS estimators can be written in terms of x_i 's and Y_i 's as

$$\begin{cases} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} Y_i \\ \hat{\beta}_0 &= \bar{Y} - \bar{x}\hat{\beta}_1 = \sum_{i=1}^n \left(\frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) Y_i \end{cases}$$

Note that the estimators are linear combinations of the Y_i 's, which are independent random variables before taking observations.

$$\begin{cases} \hat{\beta}_1 &= \sum_{i=1}^n c_i Y_i, & c_i &= \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ \hat{\beta}_0 &= \sum_{i=1}^n d_i Y_i, & d_i &= \frac{1}{n} - \bar{x}c_i \end{cases}$$

The linear combination coefficient c_i 's have the following useful properties.

$$\sum_{i=1}^n c_i = 0, \quad \sum_{i=1}^n c_i x_i = 1, \quad \sum_{i=1}^n c_i^2 = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Unbiasedness of estimators $\hat{\beta}_0, \hat{\beta}_1$

Since Y_i 's are random variables (before observing the data), so are the estimators $\hat{\beta}_0, \hat{\beta}_1$, and they are unbiased estimators of the parameters β_0, β_1 .

$$\mathbb{E}(\hat{\beta}_1) = \sum_{i=1}^n c_i \mathbb{E}(Y_i) = \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i = \beta_1$$

$$\mathbb{E}(\hat{\beta}_0) = \mathbb{E}(\bar{Y} - \bar{x}\hat{\beta}_1) = (\beta_0 + \bar{x}\beta_1) - \bar{x}\beta_1 = \beta_0$$

Variances and covariance of estimators $\hat{\beta}_0, \hat{\beta}_1$

Use the linear property of the estimator,

$$Var(\hat{\beta}_1) = \sum_{i=1}^n Var(c_i Y_i) = \sum_{i=1}^n c_i^2 Var(Y_i) = \sum_{i=1}^n c_i^2 \sigma^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Apply the result for $Var(\hat{\beta}_1)$ to the variance calculation for $\hat{\beta}_0$,

$$Var(\hat{\beta}_0) = Var(\bar{Y} - \bar{x}\hat{\beta}_1) = Var(\bar{Y}) + \bar{x}^2 Var(\hat{\beta}_1) - 2\bar{x} Cov(\bar{Y}, \hat{\beta}_1) = \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

In the last step, we use the result

$$Cov(\bar{Y}, \hat{\beta}_1) = \sum_{i=1}^n c_i Cov(\bar{Y}, Y_i) = \sum_{i=1}^n c_i Cov(\frac{1}{n} Y_i, Y_i) = \sum_{i=1}^n c_i \frac{\sigma^2}{n} = 0.$$

Finally, the covariance of the LS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ can be derived.

$$\begin{aligned} Cov(\hat{\beta}_0, \hat{\beta}_1) &= Cov(\bar{Y} - \bar{x}\hat{\beta}_1, \hat{\beta}_1) \\ &= Cov(\bar{Y}, \hat{\beta}_1) - Cov(\bar{x}\hat{\beta}_1, \hat{\beta}_1) \\ &= 0 - \bar{x} Cov(\hat{\beta}_1, \hat{\beta}_1) = \frac{-\bar{x} \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

Remarks

- The accuracy or precision of an estimator is reciprocal to its variance.

The formula for $Var(\hat{\beta}_1)$ can be written as

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{n} \cdot \frac{1}{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

This expression indicates that:

- The variance of $\hat{\beta}_1$ is proportional to the noise level, measured by σ^2 .
The lower the noise level, the more accurate of the estimation of the slope β_1 .
- The variance of $\hat{\beta}_1$ is inversely proportional to the sample size n .
The large the sample size, the better the precision of the estimation of β_1 .
- The variance of $\hat{\beta}_1$ is inversely proportional to the spread of the x_i 's, measured by $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$.
Hence, with the same measurement scale and same number of observations, the x 's should be chosen to cover a wider range in order to get more accurate estimate of the slope β_1 .

- The formula for $Var(\hat{\beta}_0)$

$$Var(\hat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

also shows the inverse relationship of the variance of $\hat{\beta}_0$ and the spread of the x_i 's.

In addition, the location of the mean $\bar{x}$ has an effect on the intercept β_0 . The closer the $\bar{x}$ to 0, the smaller the variance in the estimation of the intercept β_0 .

- The covariance $Cov(\hat{\beta}_0, \hat{\beta}_1)$ demonstrates that

If $\bar{x} = 0$, $\hat{\beta}_0, \hat{\beta}_1$ are uncorrelated.

If $\bar{x} > 0$, $\hat{\beta}_0, \hat{\beta}_1$ are negatively correlated.

If $\bar{x} < 0$, $\hat{\beta}_0, \hat{\beta}_1$ are positively correlated.

Need and usage of the normality assumption

Assumptions on ε_i 's

Note that, the only properties of the error term ε_i used in the above derivations are that ε_i 's are i.i.d. with mean 0 and variance σ^2 . Normal distribution assumption of the error term has not been used.

To obtain statistical inference, such as conducting hypothesis testing or constructing confidence intervals on the LS estimators, distribution assumptions are needed.

Joint distribution of estimators $\hat{\beta}_0, \hat{\beta}_1$

Assume that the errors ε_i are i.i.d. $\sim N(0, \sigma^2)$. That is, we assume

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n,$$

Then Y_i 's are independent (but not identically distributed) normal random variables, and

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Recall that LS estimators

$$\hat{\beta}_1 = \sum_i c_i Y_i, \quad \hat{\beta}_0 = \sum_i d_i Y_i$$

are linear combinations of the Y_i 's. Consequently, $\hat{\beta}_0, \hat{\beta}_1$ are of joint normal.

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} \sim N_2 \left(\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \begin{bmatrix} \frac{1}{n} \sum_{i=1}^n x_i^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix} \right)$$

where we applied another expression for $Var(\hat{\beta}_0)$,

$$Var(\hat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{n} \cdot \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

MLE for the coefficients and σ^2

With the normality assumption, maximum likelihood estimators can be obtained for the parameters in the linear model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n.$$

which is equivalent to

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2), \quad Y_i \text{'s are mutually independent, } i = 1, \dots, n.$$

Their joint density under the linear model is the likelihood function

$$L = L(\beta_0, \beta_1, \sigma^2) = L(\beta_0, \beta_1, \sigma^2 \mid (x_i, y_i), i = 1, \dots, n) = f(y_1, \dots, y_n) \\ = \prod_{i=1}^n f_i(y_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} (y_i - (\beta_0 + \beta_1 x_i))^2}$$

Then the log-likelihood function for the linear regression model is

$$\log L = \log_e L(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

The MLE for β_1, β_0 and σ^2 should maximize the likelihood,

$$\{\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2\} = \arg \max_{\beta_0, \beta_1, \sigma^2} \log L(\beta_0, \beta_1, \sigma^2)$$

To derive the MLE, taking partial derivatives and maximizing over β_0, β_1 ,

$$\begin{cases} \frac{\partial}{\partial \beta_0} \log L(\beta_0, \beta_1, \sigma^2) = \left(-\frac{1}{2\sigma^2}\right) \sum_{i=1}^n 2(y_i - (\beta_0 + \beta_1 x_i))(-1) = 0 \\ \frac{\partial}{\partial \beta_1} \log L(\beta_0, \beta_1, \sigma^2) = \left(-\frac{1}{2\sigma^2}\right) \sum_{i=1}^n 2(y_i - (\beta_0 + \beta_1 x_i))(-x_i) = 0 \\ \frac{\partial}{\partial \sigma^2} \log L(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 = 0 \end{cases} \quad (2)$$

The first two equations in (2) by the ML method are equivalent to the ones in (1) from the LS derivation. From these two equations we can derive the unique critical values

$$\hat{\beta}_0 = \bar{Y} - \bar{x}\hat{\beta}_1, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Further checks can verify that these are the maximizers of the likelihood function, as we did for the LSE. Thus the ML estimators for β_0, β_1 are the same as the LS estimators.

Note that the above $\hat{\beta}_0, \hat{\beta}_1$ maximize the likelihood function for any values of σ^2 . We can express this as

$$\{\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2\} = \arg \max_{\beta_0, \beta_1, \sigma^2} \log L(\beta_0, \beta_1, \sigma^2) = \arg \max_{\sigma^2} \log L(\hat{\beta}_0, \hat{\beta}_1, \sigma^2)$$

Therefore we only need to consider the last step, which is an one-dimensional optimization problem of finding MLE for σ^2 at given $(\hat{\beta}_0, \hat{\beta}_1)$.

To derive the ML estimator for σ^2 , denote the sum of squares of the estimated errors or residuals as

$$RSS = \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 = SSE = SS_E$$

Evaluate the third equation in (2) at the ML estimators $\hat{\beta}_0, \hat{\beta}_1$,

$$\frac{\partial \log L(\hat{\beta}_0, \hat{\beta}_1, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{SS_E}{2\sigma^4} = 0 \quad \Rightarrow \quad \sigma^2 = \frac{1}{n} SS_E.$$

One should check quickly that this critical point is indeed the maximizer. Let's apply the first derivative test by checking the signs of the first derivative at the critical point.

$$\frac{\partial \log L(\hat{\beta}_0, \hat{\beta}_1, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{SS_E}{2\sigma^4} \quad \begin{cases} > 0 \\ = 0 \\ < 0 \end{cases} \quad \text{when} \quad \sigma^2 \begin{cases} < \\ = \\ > \end{cases} \frac{1}{n} SS_E$$

The first derivative test above shows that $\sigma^2 = \frac{1}{n} SS_E$ achieves the unique maximum of the log-likelihood function.

$$\Rightarrow \quad \hat{\sigma}_{ML}^2 = \frac{1}{n} SS_E$$

Remarks

- From the normality assumption of the linear model, it follows that (to be proved later using matrix algebra)

$$\frac{SS_E}{\sigma^2} = \frac{\sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2}{\sigma^2} \sim \chi_{n-2}^2$$

The reduction of d.f. is due to the two estimators which are estimated from the data.

- Consequently, the ML estimator of σ^2 is biased,

$$\mathbb{E}(\hat{\sigma}_{ML}^2) = \mathbb{E}\left(\frac{1}{n} SS_E\right) = \frac{n-2}{n} \sigma^2$$

- Thus, the unbiased estimator

$$s^2 = \frac{1}{n-2} SS_E$$

is often used as an estimator for σ^2 .

- Goodness of estimators depend on the judging criteria, e.g., unbiased-ness, variance, mean squared error.

Even though s_2 is unbiased, the MLE estimator has a slightly smaller variance, therefore the MLE $\hat{\sigma}_{ML}^2$ is not necessarily inferior than the unbiased estimator s^2 .

Prediction for the mean of new observations

Suppose that a line is fit to n points under the line model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n.$$

Given any additional new data point at $x = x_0$, the model estimated expected response is on the fitted line,

$$\widehat{\mathbb{E}(Y_0)} = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

As for any estimation, we should know its expected value and variance, as well as its confidence intervals.

By the unbiasedness of $\hat{\beta}_0$ and $\hat{\beta}_1$,

$$\mathbb{E}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \beta_0 + \beta_1 x_0$$

To obtain the variance, recall

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad Var(\hat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad Cov(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{x} \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Then the variance of the predicted mean $\hat{\beta}_0 + \hat{\beta}_1 x_0$ is

$$\begin{aligned} \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) &= \text{Var}(\hat{\beta}_0) + x_0^2 \text{Var}(\hat{\beta}_1) + 2x_0 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2 \bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sigma^2 \bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} + (\bar{x}^2 + x_0^2 - 2x_0 \bar{x}) \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} (\bar{x}^2 + x_0^2 - 2x_0 \bar{x}) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \end{aligned}$$

Remarks

As a function of x_0 , the curve of the variance $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0)$ resembles an upward quadratic function with vertex at $(\bar{x}, \sigma^2/n)$. This implies that, the accuracy of the estimator $\hat{\beta}_0 + \hat{\beta}_1 x_0$ depends on the (squared) distance between the new x_0 and the sample mean $\bar{x}$ of the n data points used to fit the model.

- The closer to the center $\bar{x}$, the more accurate the estimate of $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.
- The farther away from the center $\bar{x}$, the less certain of the estimate of $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

The derivation of the mean and variance gives us the normality of the estimated response $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$,

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \sim N \left(\beta_0 + \beta_1 x_0, \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \right)$$

Equivalently,

$$\frac{\hat{\beta}_0 + \hat{\beta}_1 x_0 - (\beta_0 + \beta_1 x_0)}{\sigma \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}} \sim N(0, 1)$$

Thus a $(1 - \alpha)\%$ confidence interval for the true mean $\beta_0 + \beta_1 x_0$ is

$$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm z_{\alpha/2} \sigma \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

Note that σ^2 is involved, which is unknown. To obtain a confidence interval for the true mean $\beta_0 + \beta_1 x_0$ using data only, we use the fact that (to be proved later)

- $\frac{RSS}{\sigma^2} = \frac{\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{\sigma^2} \sim \chi_{n-2}^2$
- $RSS \perp \left[\begin{array}{c} \hat{\beta}_0 \\ \hat{\beta}_1 \end{array} \right]$

Consequently, if we replace σ^2 by the unbiased estimator

$$s^2 = \frac{\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{n-2} = \frac{RSS}{n-2}, \quad \text{with } \mathbb{E}(s^2) = \sigma^2,$$

and since s^2 is independent of $\hat{\beta}_0 + \hat{\beta}_1 x_0$, we have a t statistic,

$$\frac{\hat{\beta}_0 + \hat{\beta}_1 x_0 - (\beta_0 + \beta_1 x_0)}{s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}} = \frac{\frac{\hat{\beta}_0 + \hat{\beta}_1 x_0 - (\beta_0 + \beta_1 x_0)}{\sigma \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}}}{\sqrt{\frac{RSS}{\sigma^2} / (n-2)}} \sim t_{n-2}$$

A $(1 - \alpha)\%$ confidence interval for the mean $\beta_0 + \beta_1 x_0$ is

$$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{n-2, 1-\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

where the t -quantile is defined by $\mathbb{P}(t_{n-2} \geq t_{n-2, 1-\alpha/2}) = \alpha/2$ as usual, and

$$s = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{n-2}} = \sqrt{\frac{RSS}{n-2}}$$

- As x_0 varies across all values, the confidence intervals form a curved confidence band flanking the fitted regression line.
- The width of the confidence band is the narrowest when x_0 is at $\bar{x}$, and the confidence band gets wider as $|x_0 - \bar{x}|$ increases.

Model selection t -test

Various t tests can be conducted to test hypotheses of interest.

Example

Does x explain the variation of Y at all? In other word, should x_i be in the model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$?

This can be checked by a hypothesis test

$$\begin{cases} H_o : \beta_1 = 0 \\ \beta_a : \beta_1 \neq 0 \end{cases}$$

We have already derived that

$$\hat{\beta}_1 \sim N \left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$$

thus

$$\frac{\hat{\beta}_1 - \beta_1}{\sigma \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \sim N(0, 1)$$

Replacing σ^2 with the estimator $s^2 = RSS/(n-2)$ (which is independent of $\hat{\beta}_1$) as in the above prediction confidence band derivation, we have

$$\frac{\hat{\beta}_1 - \beta_1}{s \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \sim t_{n-2}$$

Under the null $H_o : \beta_1 = 0$, the test statistic

$$T_0 = \frac{\hat{\beta}_1}{s \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \sim t_{n-2}$$

At test level α , we reject H_o when

$$|T_0| > t_{n-2, 1-\alpha/2}$$

Confidence intervals can also be constructed using t distributions accordingly.